SkyNow Weather App: Project Report

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1. **What was done**

SkyNow is a weather application built with plain HTML, CSS and JavaScript, without any framework. The user can search for a city or use their GPS location.

One of the main features of my application is that it uses three weather providers: Open-Meteo, MET Norway, and OpenWeatherMap. Users can see the available weather information from these providers and compare their data.

The application also comes with an hourly forecast of the next 24 hours, a 7 days forecast and a temperature comparison chart. Users can save their favorite places and can open them up later. They can also switch the unit of temperature between Celsius, Fahrenheit and Kelvin.

Also added styling to the app based on the weather conditions. As the weather changes, and depending on the time of day, the appearance of the page changes. The layout has been designed to scale to various screen sizes including mobile.

2. **Tools and data sources**
   - **HTML:** To create the structure of the application.
   - **CSS:** To design the page and make the layout responsive.
   - **JavaScript:** To handle user interactions, retrieve weather data, and update the page.
   - **Weather providers:** Open-Meteo, MET Norway (Norwegian Meteorological Institute), OpenWeatherMap
   - **Open-Meteo Geocoding API** for city search
   - **Libraries:** Chart.js (graph), Font Awesome (icons), Google Fonts (Manrope)
   - **Browser APIs:** Geolocation API, localStorage (favorites)
   - **Visual Studio Code:** To write and organize the code.

Project structure: The project folder weather-app contains index.html, css/style.css, js/script.js and report.pdf.

3. **Points requested and justification**

The following table explains the features I implemented and the points I am requesting according to the course requirements.

Feature	Points	Justification

User can search for locations	1	Search box uses the Open-Meteo geocoding API; the user picks a place from the results
GPS location	2	The “Locate” button uses the Geolocation API
Two providers	3	Open-Meteo and MET Norway are separate services
Three providers	2	OpenWeatherMap is used as a third service
Current weather	1	Current temperature and details are shown for each provider
24-hour forecast	3	Hourly cards with time, icon and temperature
7-day forecast	3	Daily list with icon and min/max temperature
Icons	3	Font Awesome icons with numbers in all forecast elements
Look reflects the weather	2	The page is dark at night and changes from blue (cold) to green, orange and red (hot)
Graph with several forecasts	3	One chart shows the hourly temperatures of all three providers
Favorites	2	A Save button stores places in localStorage; they appear as quick links
°C / °F / K	2	Unit switch in the top bar, the data is converted without a new API call
Responsive	4	Layout adapts to desktop, tablet and mobile, tested in desktop, tablet and mobile sizes using the browser developer tools (F12).
Works in Firefox, Safari, Edge, Chrome	3	Tested in: Chrome, Safari and Firefox
Clear directory structure	2	Files are organized into separate folders: index.html is in the project root, styles are in css/style.css, and the JavaScript is in js/script.js. HTML, CSS and JavaScript are kept in separate files, and the report is in the same project folder as report.pdf.
Report	3	This document: it describes what was done, the tools used, the requested points with justification, a user manual and the declaration of AI usage
Own feature	1	Additional provider-specific weather information is displayed in the application, and when there are provider failures, the application will deal with it where it can.
Total points requested	**40**	The final score depends on the course assessment.

Own feature (1 point): I expanded the application with additional information to make it more useful. Information from each provider, including feels-like temperature, UV index, cloud cover and humidity, can be displayed according to the information available. The application is also showing the time of the sunrise/sunset. One more helpful feature is that it can still display the available weather data even if one of the providers fails, without relying solely on a single provider. I think these features will help users to see the weather information from different sources in a more easily comparable way.

4. **User manual**

Users can search for a city and choose a location from the search results or click Locate and let the location be accessed to get the local weather. They can save locations by using Save and reopen them from the Favorites. Temperature can be converted from Celsius to Fahrenheit or Kelvin and vice versa. The hourly and 7-days forecasts can also be displayed and temperatures can be compared in the chart. The hourly forecast may be scrolled horizontally on smaller screens.

5. **Testing and Limitations**

I tested the application locally and checked its main features. Prior to submission, I will check the search, location access, temperature conversion, faves, forecast, and comparison chart. This application needs to be connected to the Internet to get the weather information. Certain data may not be available if no provider is found. The Locate feature also needs user's consent.

Browsers tested: Based on the instructions given in the course, the API key for OpenWeatherMap is in the file `script.js` and will be removed after the project is accepted.

6. **Declaration of AI usage**

I used Claude (Anthropic) and ChatGPT (OpenAI) to help me understand the project requirements, learn how to use weather APIs, troubleshoot issues, and prepare the first draft of this report. I was initially confused about how to integrate the three weather providers, and the AI tools helped me understand the process.

The application is self implemented and tested. I have also compared the features implemented to the grading criteria and reviewed and edited the report. I have utilized AI tools to provide guidance, support, and feedback, but not to replace personal understanding and work.